

21. Event Link Controller (ELC)

21.1 Overview

The event link controller (ELC) uses the interrupt requests generated by various peripheral modules as event signals, and interconnects (links) peripheral modules. As a result, peripheral modules can directly perform interlinked operation among them without using software.

Event signals can be output regardless of the settings of the corresponding interrupt request enable bits.

Table 21.1 lists the specifications of the ELC, and Figure 21.1 shows a block diagram of the ELC.

Table 21.1 ELC Specifications

Item	Description
Event link function	135 types of event signals can be directly interconnected to modules. - Operation for timer modules when inputting an event signal can be selected. - Event linkage operation is possible for port B and port E. - Single port*1: Event linkage operation can be set in a single specified port. - Port group*1: Event linkage operation can be set by grouping multiple specified ports among total of eight ports.
Low power consumption function	Module stop state can be set.

Note 1. When an input signal to a corresponding pin changes, an event is generated in a single port or in a port group specified as the input.

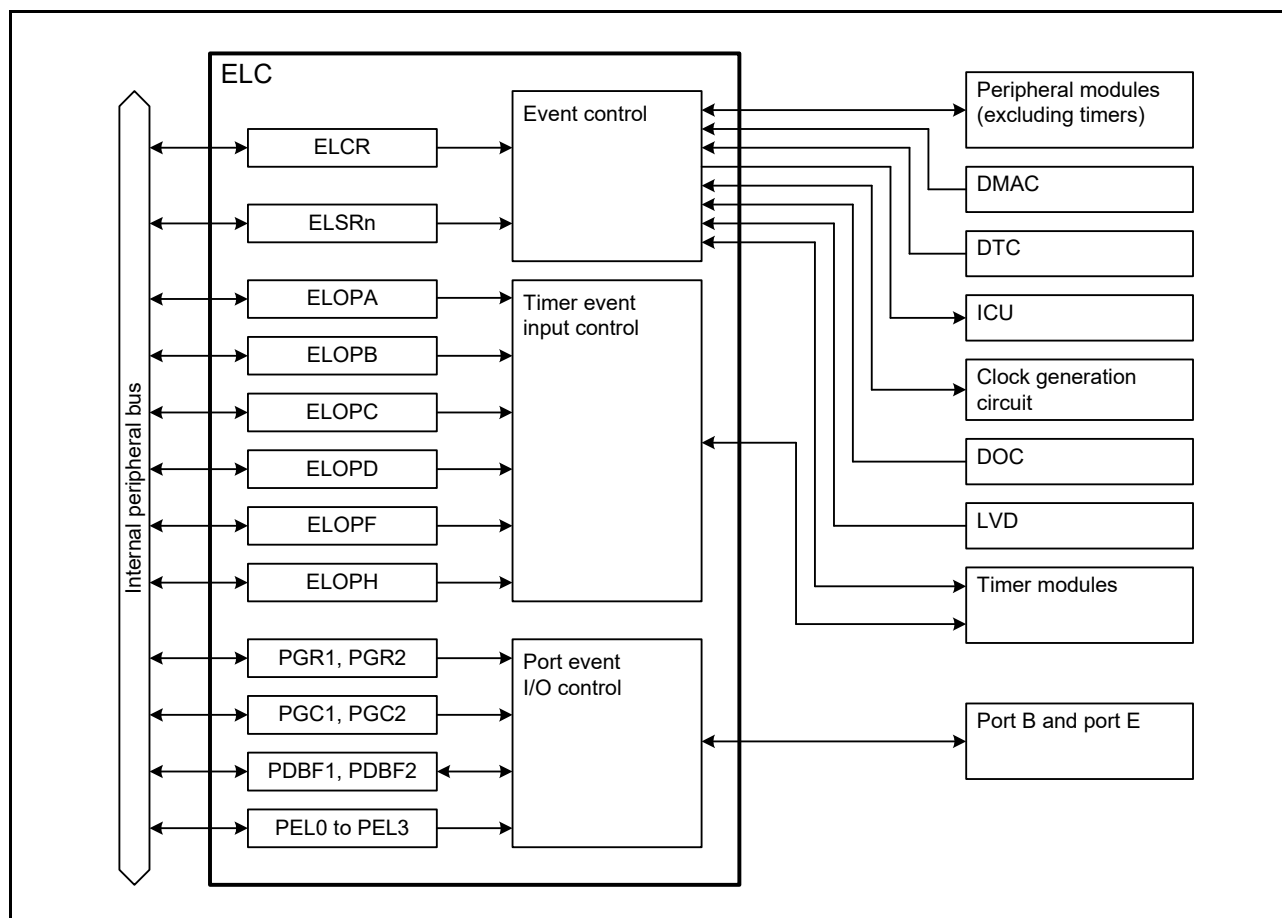


Figure 21.1 ELC Block Diagram (n = 0, 3, 4, 7, 10 to 13, 15, 16, 18 to 28, 33, 35 to 38, 45, 48 to 57)

21.2 Register Descriptions

21.2.1 Event Link Control Register (ELCR)

Address(es): ELC.ELCR 0008 B100h

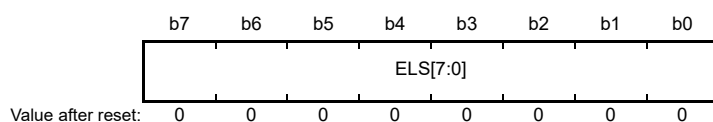
	b7	b6	b5	b4	b3	b2	b1	b0
	ELCON	—	—	—	—	—	—	—
Value after reset:	0	1	1	1	1	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b6 to b0	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b7	ELCON	All Event Link Enable	0: ELC function is disabled. 1: ELC function is enabled.	R/W

The ELCR register controls operation of the ELC.

21.2.2 Event Link Setting Register n (ELSRn) (n = 0, 3, 4, 7, 10 to 13, 15, 16, 18 to 28, 33, 35 to 38, 45, 48 to 57)

Address(es): ELC.ELSR0 0008 B101h, ELC.ELSR3 0008 B104h, ELC.ELSR4 0008 B105h, ELC.ELSR7 0008 B108h, ELC.ELSR10 0008 B10Bh, ELC.ELSR11 0008 B10Ch, ELC.ELSR12 0008 B10Dh, ELC.ELSR13 0008 B10Eh, ELC.ELSR15 0008 B110h, ELC.ELSR16 0008 B111h, ELC.ELSR18 0008 B113h, ELC.ELSR19 0008 B114h, ELC.ELSR20 0008 B115h, ELC.ELSR21 0008 B116h, ELC.ELSR22 0008 B117h, ELC.ELSR23 0008 B118h, ELC.ELSR24 0008 B119h, ELC.ELSR25 0008 B11Ah, ELC.ELSR26 0008 B11Bh, ELC.ELSR27 0008 B11Ch, ELC.ELSR28 0008 B11Dh, ELC.ELSR33 0008 B131h, ELC.ELSR35 0008 B133h, ELC.ELSR36 0008 B134h, ELC.ELSR37 0008 B135h, ELC.ELSR38 0008 B136h, ELC.ELSR45 0008 B13Dh, ELC.ELSR48 0008 B146h, ELC.ELSR49 0008 B147h, ELC.ELSR50 0008 B148h, ELC.ELSR51 0008 B149h, ELC.ELSR52 0008 B14Ah, ELC.ELSR53 0008 B14Bh, ELC.ELSR54 0008 B14Ch, ELC.ELSR55 0008 B14Dh, ELC.ELSR56 0008 B14Eh, ELC.ELSR57 0008 B14Fh



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	ELS[7:0]	Event Link Select	00h: Event signal output to the corresponding peripheral module is disabled. 01h to CDh: Set the number for the event signal to be linked. Settings other than above are prohibited.	R/W

The ELSRn register specifies an event signal to be linked to for each peripheral module. Table 21.2 shows the correspondence between the ELSRn register and the peripheral modules. Table 21.3 shows the correspondence between values set in the ELSRn register and event signals.

Table 21.2 Correspondence between the ELSRn Register and the Peripheral Modules

Register Name	Peripheral Module
ELSR0	MTU0
ELSR3	MTU3
ELSR4	MTU4
ELSR7	CMT1
ELSR10	TMR0
ELSR11	TMR1
ELSR12	TMR2
ELSR13	TMR3
ELSR15	S12AD (ELCTRG00N)
ELSR16	DA0
ELSR18	ICU (Interrupt 1)*1
ELSR19	ICU (Interrupt 2)*1
ELSR20	Output port group 1
ELSR21	Output port group 2
ELSR22	Input port group 1
ELSR23	Input port group 2
ELSR24	Single port 0
ELSR25	Single port 1
ELSR26	Single port 2
ELSR27	Single port 3
ELSR28	Clock source switching to LOCO
ELSR33	CMTW0
ELSR35	TPU0
ELSR36	TPU1
ELSR37	TPU2
ELSR38	TPU3
ELSR45	S12AD1 (ELCTRG10N)
ELSR48	GPTW event source A (common to all channels)
ELSR49	GPTW event source B (common to all channels)
ELSR50	GPTW event source C (common to all channels)
ELSR51	GPTW event source D (common to all channels)
ELSR52	GPTW event source E (common to all channels)
ELSR53	GPTW event source F (common to all channels)
ELSR54	GPTW event source G (common to all channels)
ELSR55	GPTW event source H (common to all channels)
ELSR56	S12AD (ELCTRG01N)
ELSR57	S12AD1 (ELCTRG11N)

Note 1. Specify an event number from among 63h to 6Ah. Do not set other values.

Table 21.3 Correspondence between Values Set in ELSRn.ELS[7:0] Bits and Event Signals (1/4)

ELS[7:0] Bit Value	Peripheral Modules	Event Signal Set in ELSRn
01h	Multifunction timer pulse unit 3	MTU0 compare match 0A
02h		MTU0 compare match 0B
03h		MTU0 compare match 0C
04h		MTU0 compare match 0D
05h		MTU0 compare match 0E
06h		MTU0 compare match 0F
07h		MTU0 overflow
10h		MTU3 compare match 3A
11h		MTU3 compare match 3B
12h		MTU3 compare match 3C
13h		MTU3 compare match 3D
14h		MTU3 overflow
15h		MTU4 compare match 4A
16h		MTU4 compare match 4B
17h		MTU4 compare match 4C
18h		MTU4 compare match 4D
19h		MTU4 overflow
1Ah		MTU4 underflow
1Fh	Compare match timer	CMT1 compare match 1
22h	8-bit timers	TMR0 compare match A0
23h		TMR0 compare match B0
24h		TMR0 overflow
25h		TMR1 compare match A1
26h		TMR1 compare match B1
27h		TMR1 overflow
28h		TMR2 compare match A2
29h		TMR2 compare match B2
2Ah		TMR2 overflow
2Bh		TMR3 compare match A3
2Ch		TMR3 compare match B3
2Dh		TMR3 overflow
2Eh	Realtime clock	RTC periodic event (select 1/256, 1/128, 1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1, or 2 seconds)
31h	Independent watchdog timer	IWDT underflow or refresh error
3Ah	Serial communications interfaces	SCI5 error (receive error or error signal detection)
3Bh		SCI5 receive data full
3Ch		SCI5 transmit data empty
3Dh		SCI5 transmit end
4Eh	I ² C-bus interface	RIIC0 communication error or event generation
4Fh		RIIC0 receive data full
50h		RIIC0 transmit data empty
51h		RIIC0 transmit end

Table 21.3 Correspondence between Values Set in ELSRn.ELS[7:0] Bits and Event Signals (2/4)

ELS[7:0] Bit Value	Peripheral Modules	Event Signal Set in ELSRn
52h	Serial peripheral interface	RSPI0 error (mode fault, overrun, underrun, or parity error)
53h		RSPI0 idle
54h		RSPI0 receive buffer full
55h		RSPI0 transmit buffer empty
56h		RSPI0 transmit end
58h	12-bit A/D converter	S12AD A/D conversion end
5Bh	Voltage detection circuit	LVD1 voltage detection
5Ch		LVD2 voltage detection
5Dh	DMA controller	DMAC0 transfer end
5Eh		DMAC1 transfer end
5Fh		DMAC2 transfer end
60h		DMAC3 transfer end
61h	Data transfer controller	DTC transfer end
62h	Clock generation circuit	Oscillation stop detection of clock generation circuit
63h	I/O ports	Input edge detection of input port group 1
64h		Input edge detection of input port group 2
65h		Input edge detection of single input port 0
66h		Input edge detection of single input port 1
67h		Input edge detection of single input port 2
68h		Input edge detection of single input port 3
69h	Event link controller	Software event
6Ah	Data operation circuit	DOC data operation condition met
6Ch	12-bit A/D converter	S12AD1 A/D conversion end
7Eh	Compare match timer W	CMTW0 compare match

Table 21.3 Correspondence between Values Set in ELSRn.ELS[7:0] Bits and Event Signals (3/4)

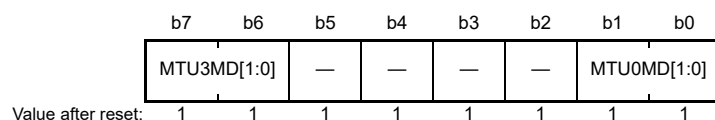
ELS[7:0] Bit Value	Peripheral Modules	Event Signal Set in ELSRn
80h	General PWM timer	GPTW0 compare match A
81h		GPTW0 compare match B
82h		GPTW0 compare match C
83h		GPTW0 compare match D
84h		GPTW0 compare match E
85h		GPTW0 compare match F
86h		GPTW0 overflow
87h		GPTW0 underflow
88h		GPTW1 compare match A
89h		GPTW1 compare match B
8Ah		GPTW1 compare match C
8Bh		GPTW1 compare match D
8Ch		GPTW1 compare match E
8Dh		GPTW1 compare match F
8Eh		GPTW1 overflow
8Fh		GPTW1 underflow
90h		GPTW2 compare match A
91h		GPTW2 compare match B
92h		GPTW2 compare match C
93h		GPTW2 compare match D
94h		GPTW2 compare match E
95h		GPTW2 compare match F
96h		GPTW2 overflow
97h		GPTW2 underflow
98h		GPTW3 compare match A
99h		GPTW3 compare match B
9Ah		GPTW3 compare match C
9Bh		GPTW3 compare match D
9Ch		GPTW3 compare match E
9Dh		GPTW3 compare match F
9Eh		GPTW3 overflow
9Fh		GPTW3 underflow
A0h	Ethernet controller	EPTPC STCA timer 0 rising edge detection
A1h		EPTPC STCA timer 1 rising edge detection
A2h		EPTPC STCA timer 2 rising edge detection
A3h		EPTPC STCA timer 3 rising edge detection
A4h		EPTPC STCA timer 4 rising edge detection
A5h		EPTPC STCA timer 5 rising edge detection
A6h		EPTPC STCA timer 0 falling edge detection
A7h		EPTPC STCA timer 1 falling edge detection
A8h		EPTPC STCA timer 2 falling edge detection
A9h		EPTPC STCA timer 3 falling edge detection
AAh		EPTPC STCA timer 4 falling edge detection
ABh		EPTPC STCA timer 5 falling edge detection

Table 21.3 Correspondence between Values Set in ELSRn.ELS[7:0] Bits and Event Signals (4/4)

ELS[7:0] Bit Value	Peripheral Modules	Event Signal Set in ELSRn
ACh	16-bit timer pulse unit	TPU0 compare match A
ADh		TPU0 compare match B
AEh		TPU0 compare match C
AFh		TPU0 compare match D
B0h		TPU0 overflow
B1h		TPU1 compare match A
B2h		TPU1 compare match B
B3h		TPU1 overflow
B4h		TPU1 underflow
B5h		TPU2 compare match A
B6h		TPU2 compare match B
B7h		TPU2 overflow
B8h		TPU2 underflow
B9h		TPU3 compare match A
BAh		TPU3 compare match B
BBh		TPU3 compare match C
BCh		TPU3 compare match D
BDh		TPU3 overflow
C6h	General PWM timer	GPTW0 A/D converter start request A
C7h		GPTW0 A/D converter start request B
C8h		GPTW1 A/D converter start request A
C9h		GPTW1 A/D converter start request B
CAh		GPTW2 A/D converter start request A
CBh		GPTW2 A/D converter start request B
CCh		GPTW3 A/D converter start request A
CDh		GPTW3 A/D converter start request B
Settings other than above are prohibited.		

21.2.3 Event Link Option Setting Register A (ELOPA)

Address(es): ELC.ELOPA 0008 B11Fh



Bit	Symbol	Bit Name	Description	R/W
b1, b0	MTU0MD[1:0]	MTU0 Operation Select	b1 b0 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Input capture* ¹ 1 1: Event output is disabled.	R/W
b5 to b2	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b7, b6	MTU3MD[1:0]	MTU3 Operation Select	b7 b6 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Input capture* ² 1 1: Event output is disabled.	R/W

Note 1. The MTU0.TCNT value is captured into the MTU0.TGRA register.

Note 2. The MTU3.TCNT value is captured into the MTU3.TGRA register.

The ELOPA register specifies the operations of MTU0 and MTU3 when an event signal is input. Set 11b (event output is disabled) when the ELC function is not used.

21.2.4 Event Link Option Setting Register B (ELOPB)

Address(es): ELC.ELOPB 0008 B120h



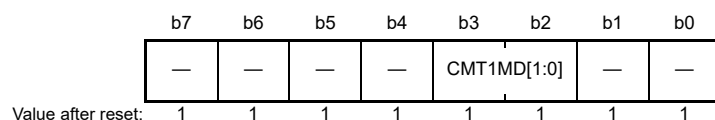
Bit	Symbol	Bit Name	Description	R/W
b1, b0	MTU4MD[1:0]	MTU4 Operation Select	b1 b0 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Input capture* ¹ 1 1: Event output is disabled.	R/W
b7 to b2	—	Reserved	These bits are read as 1. The write value should be 1.	R/W

Note 1. The MTU4.TCNT value is captured into the MTU4.TGRA register.

The ELOPB register specifies the operation of MTU4 when an event signal is input. Set 11b (event output is disabled) when the ELC function is not used.

21.2.5 Event Link Option Setting Register C (ELOPC)

Address(es): ELC.ELOPC 0008 B121h

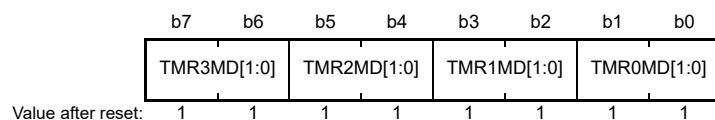


Bit	Symbol	Bit Name	Description	R/W
b1, b0	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b3, b2	CMT1MD[1:0]	CMT1 Operation Select	b3 b2 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Event counter 1 1: Event output is disabled.	R/W
b7 to b4	—	Reserved	These bits are read as 1. The write value should be 1.	R/W

The ELOPC register specifies the operation of CMT1 when an event signal is input. Set 11b (event output is disabled) when the ELC function is not used.

21.2.6 Event Link Option Setting Register D (ELOPD)

Address(es): ELC.ELOPD 0008 B122h

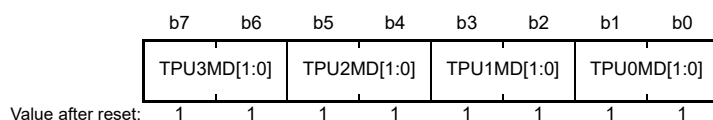


Bit	Symbol	Bit Name	Description	R/W
b1, b0	TMR0MD[1:0]	TMR0 Operation Select	b1 b0 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Event counter 1 1: Event output is disabled.	R/W
b3, b2	TMR1MD[1:0]	TMR1 Operation Select	b3 b2 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Event counter 1 1: Event output is disabled.	R/W
b5, b4	TMR2MD[1:0]	TMR2 Operation Select	b5 b4 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Event counter 1 1: Event output is disabled.	R/W
b7, b6	TMR3MD[1:0]	TMR3 Operation Select	b7 b6 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Event counter 1 1: Event output is disabled.	R/W

The ELOPD register specifies the operations of TMR0 to TMR3 when an event signal is input. Set 11b (event output is disabled) when the ELC function is not used.

21.2.7 Event Link Option Setting Register F (ELOPF)

Address(es): ELC.ELOPF 0008 B13Fh



Bit	Symbol	Bit Name	Description	R/W
b1, b0	TPU0MD[1:0]	TPU0 Operation Select	b1 b0 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Input capture* ¹ 1 1: Event output is disabled.	R/W
b3, b2	TPU1MD[1:0]	TPU1 Operation Select	b3 b2 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Input capture* ² 1 1: Event output is disabled.	R/W
b5, b4	TPU2MD[1:0]	TPU2 Operation Select	b5 b4 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Input capture* ³ 1 1: Event output is disabled.	R/W
b7, b6	TPU3MD[1:0]	TPU3 Operation Select	b7 b6 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Input capture* ⁴ 1 1: Event output is disabled.	R/W

Note 1. The TPU0.TCNT value is captured into the TPU0.TGRA register.

Note 2. The TPU1.TCNT value is captured into the TPU1.TGRA register.

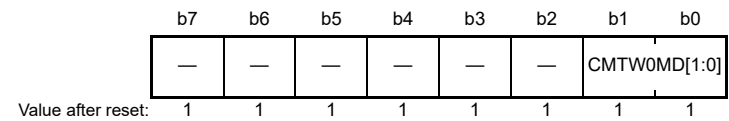
Note 3. The TPU2.TCNT value is captured into the TPU2.TGRA register.

Note 4. The TPU3.TCNT value is captured into the TPU3.TGRA register.

The ELOPF register specifies the operations of TPU0 to TPU3 when an event signal is input. Set 11b (event output is disabled) when the ELC function is not used.

21.2.8 Event Link Option Setting Register H (ELOPH)

Address(es): ELC.ELOPH 0008 B141h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	CMTW0MD[1:0]	CMTW0 Operation Select	b1 b0 0 0: Counting is started. 0 1: Counting is restarted. 1 0: Event counter 1 1: Event output is disabled.	R/W
b7 to b2	—	Reserved	These bits are read as 1. The write value should be 1.	R/W

The ELOPH register specifies the operation of CMTW0 when an event signal is input. Set 11b (event output is disabled) when the ELC function is not used.

21.2.9 Port Group Setting Register n (PGRn) (n = 1, 2)

Address(es): ELC.PGR1 0008 B123h, ELC.PGR2 0008 B124h

b7	b6	b5	b4	b3	b2	b1	b0
PGR7	PGR6	PGR5	PGR4	PGR3	PGR2	PGR1	PGR0
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	PGR0	Port Group Setting 0	0: Does not specify the port as a member of the port group. 1: Specifies the port as a member of the port group.	R/W
b1	PGR1	Port Group Setting 1		R/W
b2	PGR2	Port Group Setting 2		R/W
b3	PGR3	Port Group Setting 3		R/W
b4	PGR4	Port Group Setting 4		R/W
b5	PGR5	Port Group Setting 5		R/W
b6	PGR6	Port Group Setting 6		R/W
b7	PGR7	Port Group Setting 7		R/W

The PGRn register specifies a group of I/O ports. Among the ports, ports corresponding to bits set to 1 in the register are selected for a port group.

For example, when the PGR6 and PGR3 bits in the PGR1 register are set to 1, the PB6 and PB3 pins are selected to a port group.

Table 21.4 shows the PGRn register and corresponding ports.

Table 21.4 Registers Related to Port Groups and Corresponding Port Numbers

Port Number	Port Group Setting Register (PGRn)	Port Group Control Register (PGCn)	Port Buffer Register (PDBFn)
Port B	PGR1 register	PGC1 register	PDBF1 register
Port E	PGR2 register	PGC2 register	PDBF2 register

21.2.10 Port Group Control Register n (PGCn) (n = 1, 2)

Address(es): ELC.PGC1 0008 B125h, ELC.PGC2 0008 B126h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	PGCI[1:0]	Event Output Edge Select	b1 b0 0 0: Event signal is output upon detection of the rising edge of the input signal to the port. 0 1: Event signal is output upon detection of the falling edge of the input signal to the port. 1 x: Event signal is output upon detection of both the rising and falling edges of the input signal to the port.	R/W
b2	PGCOVE	PDBF Overwrite	0: Overwriting the PDBFn register is disabled. 1: Overwriting the PDBFn register is enabled.	R/W
b3	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b6 to b4	PGCO[2:0]	Port Group Operation Select	b6 b4 0 0 0: Low is output when an event signal is input. 0 0 1: High is output when an event signal is input. 0 1 0: The output is toggled (inverted) when an event signal is input. 0 1 1: The buffer value is output when an event signal is input. 1 x x: The output data is rotated (from MSB to LSB) in the port group when an event signal is input.	R/W
b7	—	Reserved	This bit is read as 1. The write value should be 1.	R/W

x: Don't care

For the port group set as an output, the PGCn register specifies the form of outputting the signal from the port when an event signal is input. For the port group set as an input, the PGCn register enables/disables overwriting of the PDBFn register and specifies the conditions of event generation (edge of the input signal).

Specify the I/O direction of the port by the corresponding bit in the PDR register.

Refer to Table 21.4 for the PGCn register and corresponding ports.

21.2.11 Port Buffer Register n (PDBFn) (n = 1, 2)

Address(es): ELC.PDBF1 0008 B127h, ELC.PDBF2 0008 B128h

b7	b6	b5	b4	b3	b2	b1	b0
PDBF7	PDBF6	PDBF5	PDBF4	PDBF3	PDBF2	PDBF1	PDBF0
0	0	0	0	0	0	0	0

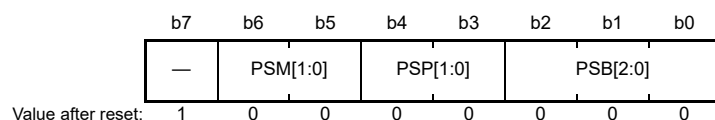
Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	PDBF0	Port Buffer 0	Specify the data to be transferred to the PODR register when an event signal is input. The setting value is valid when the PGCn.PGCO[2:0] bits are 011b or 1xxb. Write access to the bit specified as a member of the input port group is disabled. For details, refer to section 21.3, Operation.	R/W
b1	PDBF1	Port Buffer 1		R/W
b2	PDBF2	Port Buffer 2		R/W
b3	PDBF3	Port Buffer 3		R/W
b4	PDBF4	Port Buffer 4		R/W
b5	PDBF5	Port Buffer 5		R/W
b6	PDBF6	Port Buffer 6		R/W
b7	PDBF7	Port Buffer 7		R/W

The PDBFn register is an 8-bit readable/writable register used in combination with the PGRn register. Refer to section 21.3.6, I/O Port Operation When Event Signal is Input and Event Generation for the PDBFn register operations. Refer to Table 21.4 for the PDBFn register and corresponding ports.

21.2.12 Event Link Port Setting Register m (PELm) (m = 0 to 3)

Address(es): ELC.PEL0 0008 B129h, ELC.PEL1 0008 B12Ah, ELC.PEL2 0008 B12Bh, ELC.PEL3 0008 B12Ch



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	PSB[2:0]	Bit Number Specification	Set a bit number for a port to be specified as a single port.	R/W
b4, b3	PSP[1:0]	Port Number Specification	b4 b3 0 0: Setting disabled 0 1: Port B (corresponding to PGR1) 1 0: Port E (corresponding to PGR2) 1 1: Setting prohibited R/W	R/W
b6, b5	PSM[1:0]	Event Link Specification	- For the output port, specify the data to be output from the port. b6 b5 0 0: Low is output when an event signal is input. 0 1: High is output when an event signal is input. 1 x: The output is toggled (inverted) when an event signal is input. R/W - For the input port, select the edge on which the event signal is to be output. b6 b5 0 0: Event signal is output upon detection of the rising edge. 0 1: Event signal is output upon detection of the falling edge. 1 x: Event signal is output upon detection of both the rising and falling edges. R/W	R/W
b7	—	Reserved	This bit is read as 1. The write value should be 1.	R/W

x: Don't care

The PELm register specifies the single port, the operation upon an event signal input, and the conditions of event generation. This MCU can specify a total of four bits in port B and port E to respective single ports. Specify the I/O direction of the port by the corresponding bit in the PDR register.

21.2.13 Event Link Software Event Generation Register (ELSEGR)

Address(es): ELC.ELSEGR 0008 B12Dh

b7	b6	b5	b4	b3	b2	b1	b0
WI	WE	—	—	—	—	—	SEG
Value after reset: 1	0	1	1	1	1	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	SEG	Software Event Generation	0: Normal operation 1: Software event is generated.	W
b5 to b1	—	Reserved	These bits are read as 1. The write value should be 1.	R/W
b6	WE	SEG Bit Write Enable	0: Write to SEG bit is disabled. 1: Write to SEG bit is enabled.	R/W
b7	WI	ELSEGR Register Write Disable	0: Write to ELSEGR register is enabled. 1: Write to ELSEGR register is disabled.	W

The MOV instruction must be used to write to this register.

SEG Bit (Software Event Generation)

When 1 is written to this bit while the WE bit is 1, a software event is generated.

This bit is read as 0. Even if 1 is written to this bit, this bit does not become 1.

WE Bit (SEG Bit Write Enable)

The SEG bit can be written to only when the WE bit is 1.

To set this bit to 1, write 0 to the WI bit and write 1 to this bit simultaneously.

To set this bit to 0, write 0 to the WI bit and write 0 to this bit simultaneously.

WI Bit (ELSEGR Register Write Disable)

The ELSEGR register can be written to only when the value to be written to the WI bit is 0.

This bit is read as 1.

21.3 Operation

21.3.1 Relation between Interrupt Handling and Event Linking

The peripheral modules incorporated in the MCU are provided with the interrupt request status flags and the interrupt enable bits to enable/disable these interrupt requests. When an interrupt request is generated in a peripheral module, the corresponding interrupt request status flag becomes 1. If the corresponding interrupt request is enabled then, the interrupt is requested to the CPU.

In contrast, the event link controller (ELC) uses the interrupt requests generated by various peripheral modules as event signals, interconnects (links) peripheral modules, and then, makes peripheral modules perform direct interlinked operation among them without using software. Event signals can be output regardless of the setting of the corresponding interrupt enable bit.

Figure 21.2 shows the relation between the interrupt handling and ELC.

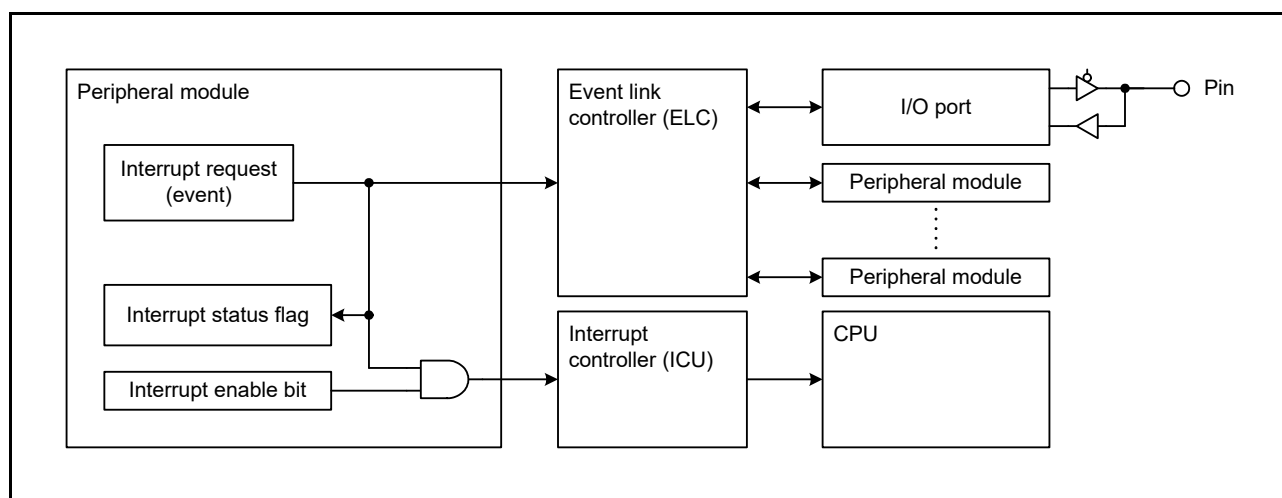


Figure 21.2 Relation between Interrupt Handling and ELC

21.3.2 Event Linkage

When events are specified in the ELSRn registers, the corresponding peripheral modules can be operated at generation of the specified events. A single peripheral module can link only with a single event. Set the ELSRn register after completing the initialization of the peripheral module to operate by an event. Table 21.5 lists the operations of peripheral modules when an event signal is input.

Table 21.5 Operations of Peripheral Modules When Event Signal is Input

Peripheral Module	Operations When Event Signal is Input		
MTU CMT CMTW TMR TPU	The following operations can be selected by setting the ELOPA to ELOPD, ELOPF, and ELOPH registers: - Starts counting when an event signal is input. - Restarts counting when an event signal is input. - Counts the input events (CMT, CMTW, TMR). - Performs input-capture operation when an event signal is input (MTU, TPU).		
GPTW	The following operations can be selected by setting registers in GPTW: - Starts counting when an event signal is input. - Stops counting when an event signal is input. - Clears counter when an event signal is input. - Increments counter when an event signal is input. - Decrements counter when an event signal is input. - Performs input-capture operation when an event signal is input.		
A/D converter	Starts A/D conversion when an event signal is input.		
D/A converter	Starts D/A conversion when an event signal is input.		
I/O ports (output)	The value of PODR register (port output data register) changes when an event signal is input (The level output from the corresponding pin changes).	Port group	- Changes the PODR register value to the specified value. - Transfers the PDBFn register value to the PODR register (n = 1, 2). - Rotates the PODR register.
		Single port	Changes the PODR register value to the specified value.
I/O ports (input)	When the signal level of the input pin changes	Port group	Generates an event.
		Single port	
	When an event signal is input	Port group	Transfers the signal level of the input pin to the PDBFn register.
		Single port	This combination cannot be used.
Clock generation circuit	Switches the clock source to the low-speed on-chip oscillator when an event signal is input.*1		
Interrupt controller	Request an interrupt to the CPU, starts DMA transfer, or starts DTC transfer when an event signal is input.		

Note 1. The SCKCR3.CKSEL[2:0] bits are modified to 000b (LOCO) regardless of the value of the protect register (PRCR.PRC0).

21.3.3 Operation of Peripheral Timer Modules When Event Signal is Input

For the timer modules except GPTW, set the ELOPA to ELOPD, ELOPE, or ELOPH register to specify the operation for when an event signal is input.

(1) Count Start Operation

When an event signal is input, the timer starts counting and the count start bit*1 in each timer control register becomes 1. An event signal that is input while the count start bit is 1 is ignored.

(2) Count Restart Operation

When an event signal is input, the timer counter is cleared. Since the count start bit*1 in each timer control register is retained, counting is restarted when an event signal is input while the count start bit is 1.

(3) Event Counter Operation

Event signal is selected as the timer count source. When an event signal is input, the timer counter is incremented.

(4) Input Capture Operation

When an event signal is input, the timer performs input-capture operation.

Note 1. Refer to the register descriptions on starting the timer in the relevant peripheral timer module section.

21.3.4 Operation of GPTW When Event Signal is Input

Eight event signals specified by the ELSR48 to ELSR55 registers are connected to every channel of GPTW as GPTW event sources A to H. To set the operation of GPTW when inputting an event signal, enable any of the event sources A to H by a corresponding bit in registers listed in Table 21.6.

Table 21.6 Operations When Event Signal is Input and Corresponding Source Select Registers

Operations on Event	Register Symbol	Register Name
Count start	GTSSR	General PWM Timer Start Source Select Register
Count stop	GTPSR	General PWM Timer Stop Source Select Register
Counter clear	GTCSR	General PWM Timer Clear Source Select Register
Count-up	GTUPSR	General PWM Timer Count-Up Source Select Register
Count-down	GTDNSR	General PWM Timer Count-Down Source Select Register
Input capture A	GTICASR	General PWM Timer Input Capture Source Select Register A
Input capture B	GTICBSR	General PWM Timer Input Capture Source Select Register B

21.3.5 Operation of A/D and D/A Converters When Event Signal is Input

When an event signal is input, the ADCSR.ADST bit and the DACR.DAOE0 bit*1 are set to 1 and the A/D and D/A converter start A/D and D/A conversion, respectively.

Note 1. Refer to the bit descriptions in the A/D converter and D/A converter sections.

21.3.6 I/O Port Operation When Event Signal is Input and Event Generation

The I/O port operation at an event signal input and conditions for event generation are set by the registers in ELC. The I/O ports that are used to set an event linkage are port B and port E.

(1) Single Ports and Port Groups

There are two event link modes: event link to single ports and event link to port groups. In the former mode, events can be interconnected to any one of the I/O ports. In the latter mode, events can be interconnected to port groups consisting of any two or more bits in the same I/O ports.

A single port can be set by the PELm.PSP[1:0] and PSB[2:0] bits ($m = 0$ to 3). A port group can be specified by setting two or more bits in the PGRn register ($n = 1, 2$) to 1. Among the ports corresponding to the bits set to 1 in the PGRn register, a port set as output becomes an output port group member, and a port set as input becomes an input port group member.

If an I/O port is specified as both a single port and a member of a port group, both functions are enabled when the corresponding port is input, whereas only the port group function is enabled when the corresponding port is output. Set the PDR register to select the direction of the I/O ports.

(2) Event Generation in Single Input Ports

A single port that is set as input generates an event signal when the input signal to the corresponding pin changes. The event generation condition is specified using the PELm.PSM[1:0] bits ($m = 0$ to 3). An example of operation is shown in Figure 21.3 (1).

(3) Single Output Ports Operation When Event Signal is Input

When an event signal is input to a single port set as output, the output level (the PODR register value) of the corresponding pin changes as specified by the PELm.PSM[1:0] bits. An example of operation is shown in Figure 21.3 (2).

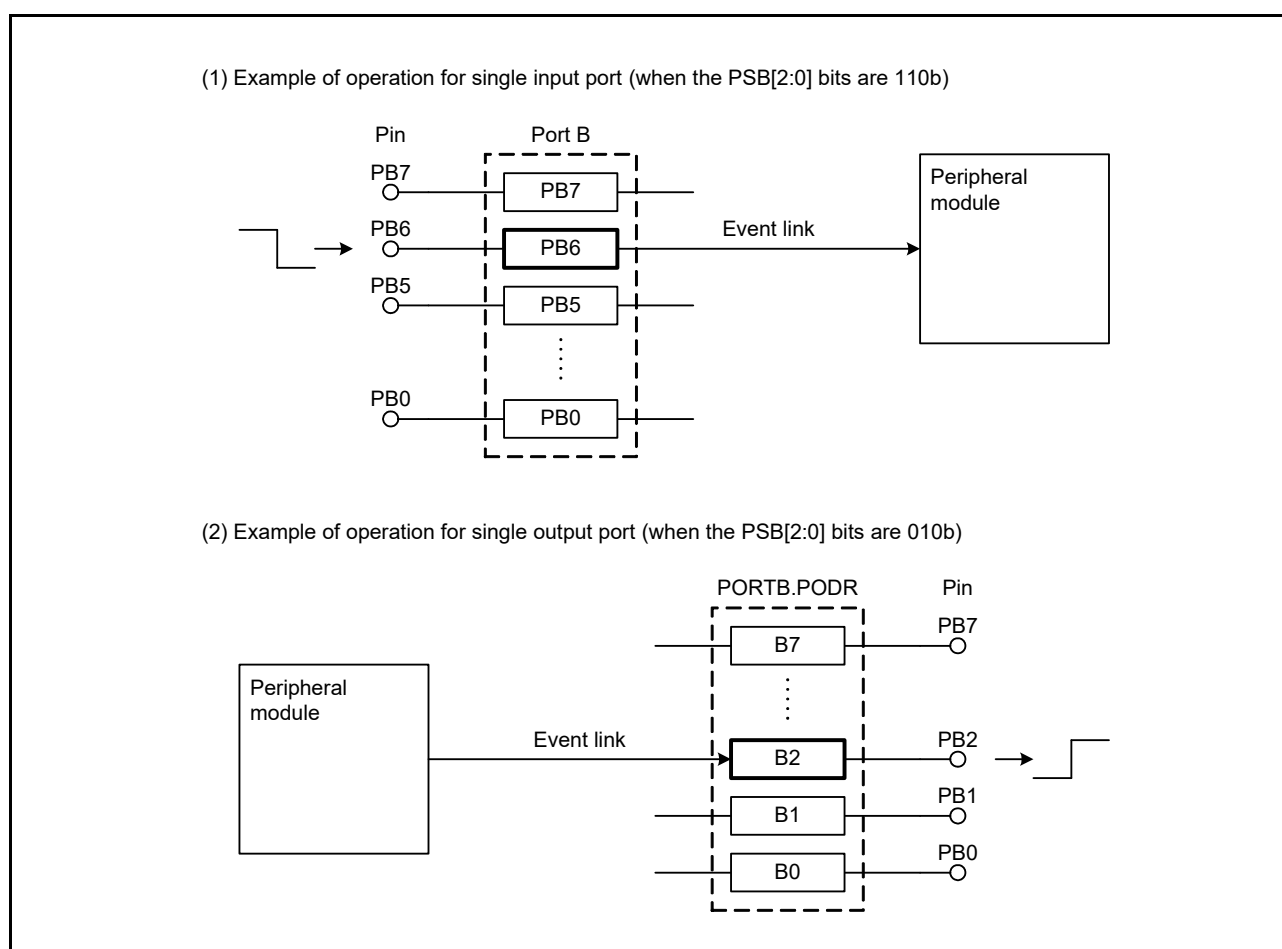


Figure 21.3 Event Linkage Related to Single Ports (Port B)

(4) Event Generation in Input Port Group

An input port group generates an event signal when any of input signals to the corresponding pins change. The event generation condition is specified using the PGCn.PGCI[1:0] bits (n = 1, 2).

(5) Input Port Group Operation When Event Signal is Input

When an event signal is input to an input port group, the level of the corresponding pins is transferred to the PDBFn register. Values of the bits corresponding to ports that are not specified as members of the input port group do not change. An example of operation is shown in Figure 21.4.

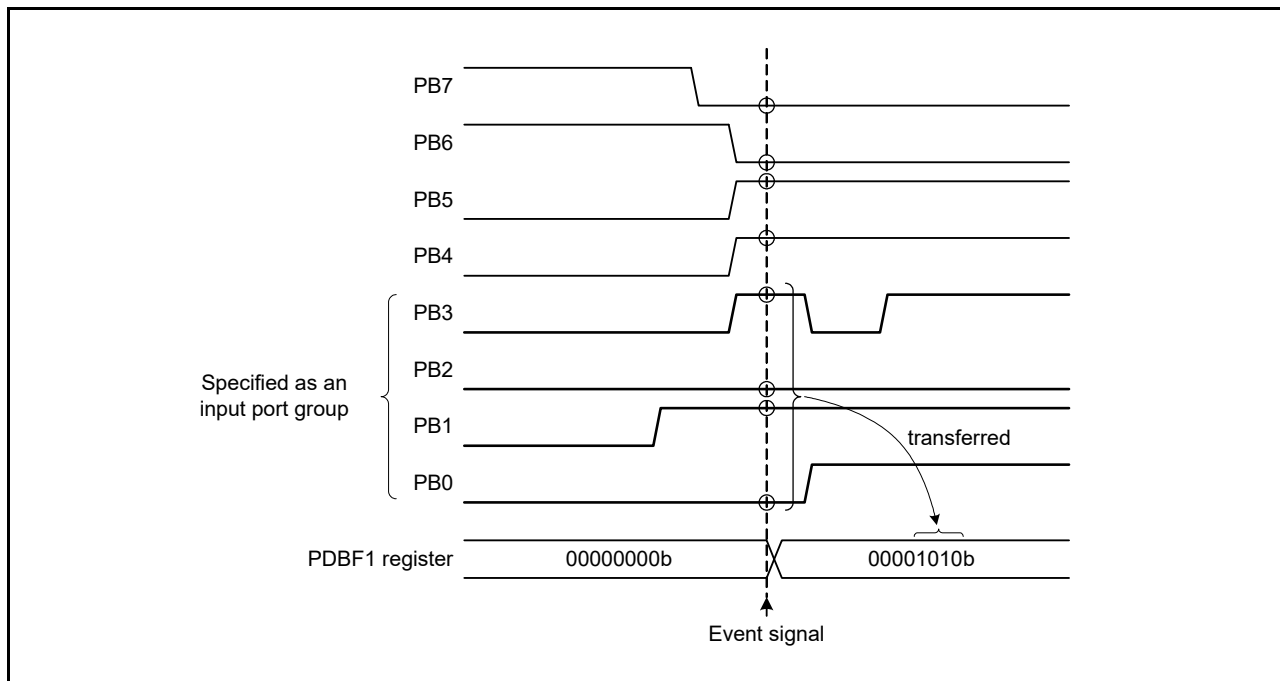


Figure 21.4 Event Linkage Related to Input Port Groups (Port B)

(6) Output Port Group Operation When Event Signal is Input

When an event signal is input to an output port group, the value of the corresponding PODR register changes according to a setting of the PGCn.PGCO[2:0] bits ($n = 1, 2$). An example of operation is shown in Figure 21.5.

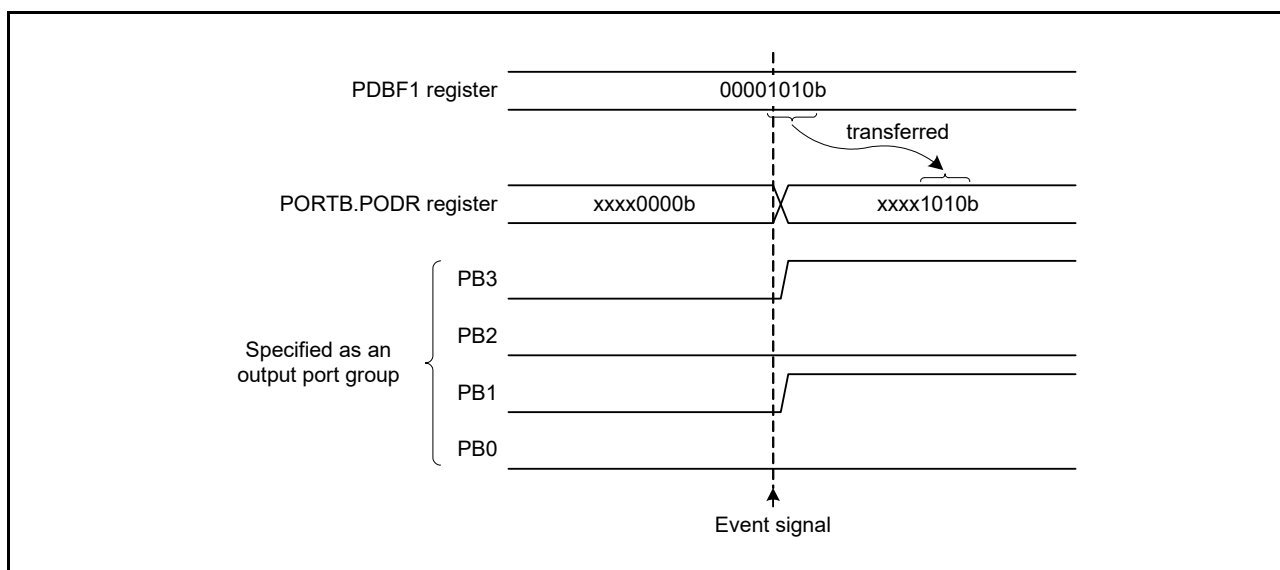


Figure 21.5 Event Linkage Related to Output Port Groups (Port B)

(7) Operation of the PDBFn Registers

(a) Input Port Groups

When an event signal is input to an input port group, the level of the corresponding pins is transferred to the PDBFn register ($n = 1, 2$). When another event signal is input to the input port group in this condition, different operations are performed depending on the PGCn.PGCOVE bit setting described as below.

- When the PGCn.PGCOVE bit is 0 (overwriting is disabled)
When the value transferred to the PDBFn register after an input of the last event signal has already been read by the CPU or DTC, the level of the corresponding pins at the time is transferred to the PDBFn register. When the value has not been read, the level of the pins is not transferred to the PDBFn register, and the input event signal is ignored.
- When the PGCn.PGCOVE bit is 1 (overwriting is enabled)
When another event signal is input to the input port group, the level of the corresponding pins is transferred to the PDBFn register.

(b) Output Port Groups

When an output port group is specified to output the PDBFn register value (PGCn.PGCO[2:0] bits = 011b), the PDBFn register value is transferred to the PODR register following an input of an event signal to the output port group. Data is not transferred to the bits corresponding to the ports that are not specified as members of the output port group.

When output data is specified to rotate in an output port group (PGCn.PGCO[2:0] bits = 1xxb), the data is transferred from the PDBFn register to the PODR register at first event signal, and the PODR register value is rotated from MSB to LSB within the relevant group at second and subsequent signals.

Examples of operation are shown in Figure 21.6.

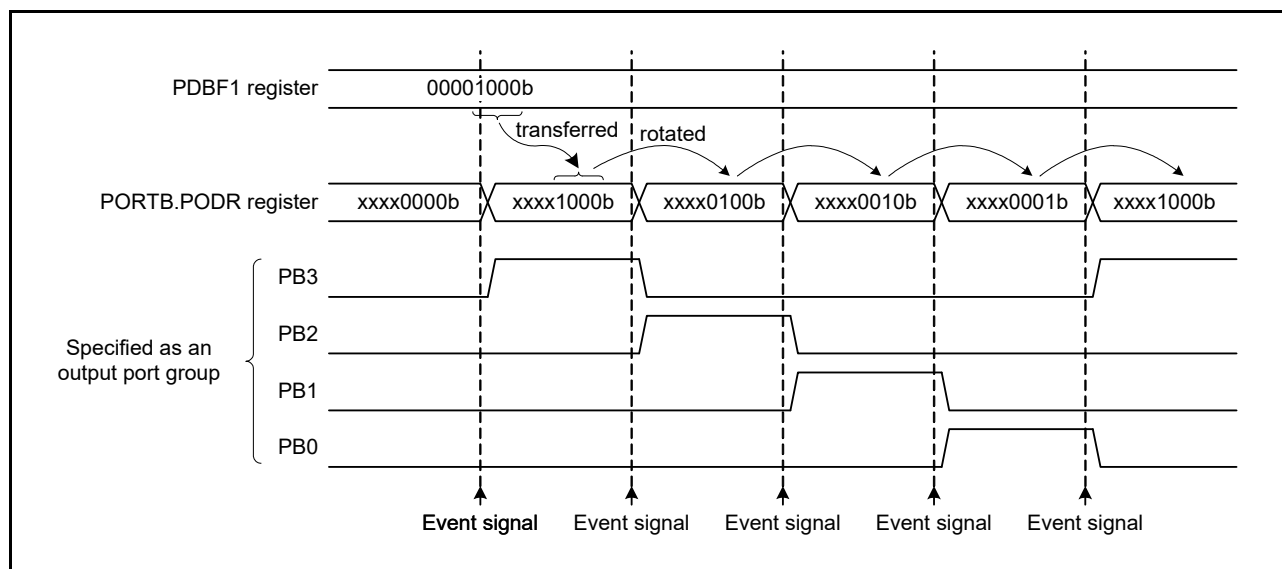


Figure 21.6 Bit-Rotating Operation of Output Port Groups (Port B)

(8) Restrictions on Writing to PODR and PDBF Registers

When the ELCR.ELCON bit is 1 (ELC function is enabled), write access to the PODR and PDBFn registers ($n = 1, 2$) becomes disabled at the following conditions.

- When a port is specified as a member of the input port group and when the event linkage is set, write access to the corresponding bit in the PDBFn register becomes disabled.
- When a port is specified as a member of the output port group, write access to the corresponding bit in the PODR register becomes disabled.
- When a port is specified as a single output port and when the event linkage for the port is set by the ELSRn register, write access to the corresponding bit in the PODR register becomes disabled.

21.3.7 Example of Procedure for Linking Events

The following describes the procedure for linking events.

- (1) Initialize the peripheral module (destination) that operates based on an event signal.
- (2) When event linkage is set to a port, set the following registers corresponding to the port.
 - PODR register: Set the initial values of the output ports.
 - PDR register: Set the I/O direction of the ports.
 - PGRn register: To operate ports for a port group, select ports to be specified as port group members ($n = 1, 2$).
 - PGCn register: Set the operation of the port group.
 - PELm register: When a port is operated as a single port, specify the port to be used, an operation of the port at an input of event signal, and the event generation condition ($m = 0$ to 3).
- (3) Set the number of the event signal to the ELSRn register corresponding to the destination peripheral module.
- (4) To link an event to a timer module, set any of the ELOPA to ELOPD, ELOPF, and ELOPH registers corresponding to the timer as required.
- (5) Set the ELCR.ELCON bit to 1, which enables linkage of all the events.
- (6) Set the operation of the peripheral module (source) from which an event signal is output, and activate the module. The preset operation of the destination peripheral module is started by the event signal that is output from the source peripheral module.
- (7) To stop event linkage of independent peripheral module, set 00h to the ELSRn register corresponding to the peripheral module. To stop linkage of all the events, set the ELCR.ELCON bit to 0.

Note: If event signal output from the RTC is to be used, make the ELC settings after the RTC settings (initialization, time setting, etc.). Unintended events may be generated if RTC settings are made after the ELC settings.

Note: When using event signal output from the LVD, set the LVD and then the ELC. Set the corresponding ELSRn register to 00h and then disable the LVD.

21.4 Usage Notes

21.4.1 Setting ELSRn Register

(1) Setting ELSR18 and ELSR19 Registers

Specify an event number from among 63h to 6Ah. Do not set the value other than preceding numbers.

21.4.2 Setting Bit-Rotating Operation of Output Port Groups

When the values of the PDBFn register ($n = 1, 2$) are changed in the bit-rotating operation mode of the output port group, set the ELSRn register again. Set intervals for generating the event as at least one PCLKB cycle when using it for bit-rotating operation.

21.4.3 Linking DMA/DTC Transfer End Signal as Event

When linking the DMA/DTC transfer end signal as an event signal, do not set the same peripheral module as the DMA/DTC transfer destination and event link destination. If set, the peripheral module might be started before DMA/DTC transfer to the peripheral module is completed.

21.4.4 Clock Settings

To link events, make sure that the ELC and the related peripheral modules are in an operational condition. The peripheral modules cannot operate if they are in the module stop state or in mode which they stop (all-module clock stop mode, software standby mode, or deep software standby mode).

21.4.5 Module Stop Function Setting

ELC operation can be disabled or enabled using module stop control register B (MSTPCRB). After reset is released, the ELC function is disabled. Register access is enabled by releasing the module stop state. For details, refer to **section 11**, Low Power Consumption.